

ie need to ~~show~~ show that for any ^{arbitrary} ~~generic~~ ~~points~~ m points, say $x_1^*, \dots, x_m^*$, $(f(x_1^*), \dots, f(x_m^*)) = \tilde{f}^*$ (say) follows a multivariate normal distr., given $\underline{Y}$, the data.

Note that showing $\tilde{f} | \underline{Y}$ is MVN is not enough., we need to show that $\tilde{f}^* | \underline{Y}$ is MVN.

Here, we can assume $\{x_1, \dots, x_n\} \cap \{x_1^*, \dots, x_m^*\} = \emptyset$ for the sake of simplicity.

~~we~~ we assume σ^2 is known.

Then,

$$\begin{aligned} \pi(\tilde{f}^* | \underline{Y}) &= \int \pi(\tilde{f}^*, \tilde{f} | \underline{Y}) d\tilde{f} = \int \pi(\tilde{f}^* | \tilde{f}, \underline{Y}) \pi(\tilde{f} | \underline{Y}) d\tilde{f} \\ &= \int \pi(\tilde{f}^* | \tilde{f}) \pi(\tilde{f} | \underline{Y}) d\tilde{f}. \end{aligned}$$

Here, ~~$\tilde{f}^* | \tilde{f} \sim \text{MVN}_{n+m}$~~

$$\begin{bmatrix} \tilde{f}^* \\ \tilde{f} \\ \underline{f} \end{bmatrix} \sim \text{MVN}_{n+m} \left(\begin{pmatrix} \mu_{\tilde{f}^*}^* \\ \mu_{\tilde{f}} \\ \mu_{\underline{f}} \end{pmatrix}, \begin{bmatrix} \Sigma_{\tilde{f}^* \tilde{f}^*}^{xx} & \Sigma_{\tilde{f}^* \tilde{f}}^* \\ \Sigma_{\tilde{f} \tilde{f}^*}^{xT} & \Sigma_{\tilde{f} \tilde{f}} \end{bmatrix} \right)$$

$$\text{Where } \mu_{\tilde{f}^*}^* = \begin{pmatrix} \mu(s_1^*) \\ \vdots \\ \mu(s_m^*) \end{pmatrix}, \Sigma_{\tilde{f}^* \tilde{f}^*}^{xx} = \begin{bmatrix} K(s_1^*, s_1^*) & \dots & K(s_1^*, s_m^*) \\ \vdots & & \vdots \\ K(s_m^*, s_1^*) & \dots & K(s_m^*, s_m^*) \end{bmatrix}$$

$$\Sigma_{\tilde{f}^* \tilde{f}}^x = \begin{bmatrix} K(s_1^*, s_1) & \dots & K(s_1^*, s_n) \\ \vdots & & \vdots \\ K(s_m^*, s_1) & \dots & K(s_m^*, s_n) \end{bmatrix}$$

$$\tilde{f}^* | \tilde{f} \sim \text{MVN}_m \left(\mu_{\tilde{f}^*}^* + \Sigma_{\tilde{f}^* \tilde{f}}^x \Sigma_{\tilde{f} \tilde{f}}^{-1} (\tilde{f} - \mu_{\tilde{f}}), \Sigma_{\tilde{f}^* \tilde{f}^*}^{xx} - \Sigma_{\tilde{f}^* \tilde{f}}^x \Sigma_{\tilde{f} \tilde{f}}^{-1} \Sigma_{\tilde{f} \tilde{f}^*}^{xT} \right)$$

Now, $\tilde{f}^* | \tilde{f} \sim \text{MVN}_m$, $\tilde{f} | \underline{Y} \sim \text{MVN}_n \Rightarrow$ by the result of convolution of MVN-MVN Gaussian being MVN, we have $\tilde{f}^* | \underline{Y} \sim \text{MVN}$.

$$\begin{aligned}
 E(\tilde{f}^* | \tilde{y}) &= E[E(\tilde{f}^* | \tilde{f}, \tilde{y})] = E\left[\tilde{\mu}^* + \Sigma^* \tilde{\Sigma}^{-1}(\tilde{f} - \tilde{\mu}) \mid \tilde{y}\right] \\
 &= \tilde{\mu}^* + \Sigma^* \tilde{\Sigma}^{-1}(E(\tilde{f} | \tilde{y}) - \tilde{\mu}) \\
 &= \tilde{\mu}^* + \Sigma^* \tilde{\Sigma}^{-1}\left(\left[\frac{1}{\sigma^2} I_n + \tilde{\Sigma}^{-1}\right]^{-1} \left[\frac{1}{\sigma^2} \tilde{y} + \tilde{\Sigma}^{-1} \tilde{f}\right] - \tilde{\mu}\right)
 \end{aligned}$$

$$\begin{aligned}
 \text{Cov}(\tilde{f}^* | \tilde{y}) &= \text{Cov}[E(\tilde{f}^* | \tilde{y}, \tilde{f})] + E[\text{Cov}(\tilde{f}^* | \tilde{y}, \tilde{f})] \\
 &= \text{Cov}\left[\tilde{\mu}^* + \Sigma^* \tilde{\Sigma}^{-1}(\tilde{f} - \tilde{\mu}) \mid \tilde{y}\right] + E\left[\Sigma^* \tilde{\Sigma}^{-1} \tilde{\Sigma}^{-1} \Sigma^{*T}\right] \\
 &= \Sigma^* \tilde{\Sigma}^{-1} \text{Cov}(\tilde{f} | \tilde{y}) \tilde{\Sigma}^{-1} \Sigma^{*T} + \Sigma^* \tilde{\Sigma}^{-1} \tilde{\Sigma}^{-1} \Sigma^{*T} \\
 &= \Sigma^* \tilde{\Sigma}^{-1} \left[\frac{1}{\sigma^2} I_n + \tilde{\Sigma}^{-1}\right]^{-1} \tilde{\Sigma}^{-1} \Sigma^{*T} + \Sigma^* \tilde{\Sigma}^{-1} \tilde{\Sigma}^{-1} \Sigma^{*T}
 \end{aligned}$$

So, for any m arbitrary points $x_1^*, \dots, x_m^*$,

$$[f(x_1^*), \dots, f(x_m^*)]' \mid \tilde{y} \sim \text{MVN}_m(E(\tilde{f}^* | \tilde{y}), \text{Cov}(\tilde{f}^* | \tilde{y}))$$

Hence, $f(\cdot) | \tilde{y} \sim \text{GP}(\tilde{\mu}(\cdot), \tilde{K}(\cdot, \cdot))$

$$\text{where } E(\tilde{f}^* | \tilde{y}) = \begin{bmatrix} \tilde{\mu}(x_1^*) \\ \vdots \\ \tilde{\mu}(x_m^*) \end{bmatrix}, \quad \text{Cov}(\tilde{f}^* | \tilde{y}) = \begin{bmatrix} \tilde{K}(x_1^*, x_1^*) & \dots & \tilde{K}(x_1^*, x_m^*) \\ \vdots & \ddots & \vdots \\ \tilde{K}(x_m^*, x_1^*) & \dots & \tilde{K}(x_m^*, x_m^*) \end{bmatrix}$$

$\Rightarrow f(\cdot) \sim \text{GP}(\mu(\cdot), K(\cdot, \cdot))$ is a Conjugate prior on $f(\cdot)$.